AI Research Intern

The AI Research Intern is an engineering role within the Program Design team, focused on building AI-powered workflows that improve how educational content is researched, created, reviewed, and validated.
You will work with AI models, prompt engineering, research automation, and software development to build tools that help instructors create accurate and high-quality learning experiences.

Task Overview

Build an AI-powered Lesson Creation Tool that generates well-researched lesson content using a Lesson Title and Learning Objective as inputs.
The tool should research relevant books, websites, and YouTube videos, synthesize the information into structured lesson content, and validate it through two independent Quality Control (QC) layers.
The complete lesson should be designed to be consumed within 45 minutes, either through self-learning or instructor-led teaching.

What You Need to Build

1. AI-Powered Lesson Generation

- Accept Lesson Title and Learning Objective as inputs.
- Research relevant information from books, websites, and YouTube videos.
- Generate structured, engaging lesson content with explanations, examples, and practical applications.
- Include relevant YouTube videos as supplementary learning resources.
- Include reference links at the end of the lesson.
- The generated lesson content must be designed to be completed within **45 minutes**, including all explanations, examples, activities, and knowledge checks.

2. Two Quality Control (QC) Layers

QC Layer 1: Research and Factual Accuracy

- Verify factual correctness and identify unsupported or misleading claims.
- Check the reliability and relevance of sources.
- Identify gaps or inconsistencies in the content.

QC Layer 2: Instructional Quality

- Check alignment with the learning objective.
- Evaluate clarity, structure, and relevance.
- Identify gaps in explanations, examples, or practical applications.

Both layers should generate separate audit reports with identified issues and suggested improvements.

3. Review, Edit, and Export

- Allow instructors to review, edit, and save lesson content.
- Allow instructors to review and apply QC recommendations.
- Provide a preview of the final lesson.
- Allow export of the finalized lesson, references, and QC reports.

4. Error Handling

- Handle empty inputs and research or AI generation failures.
- Preserve previously saved work and instructor edits.
- Allow failed steps to be retried.

Sample Inputs for Testing

Lesson Title: HTTP Requests and Responses

Learning Objective: Explain GET and POST methods and distinguish between 2xx, 4xx, and 5xx response status categories.

Additionally, test the workflow with one other lesson title and learning objective of your choice. Verify the factual accuracy of the generated content and demonstrate both QC layers.

Working Guidelines

- You are free to use any technology stack, AI model, or research tool.
- Responsible use of AI for research, coding, and debugging is encouraged.
- Ensure that you understand and can explain your implementation and technical decisions.
- Deployment is not required. A locally running workflow is sufficient.
- Focus on functionality and research quality rather than visual design.

What to Submit

1. Project Files: A public GitHub repository link or ZIP file containing the complete source code and basic setup instructions.
2. Demo Video: A video demonstrating lesson generation, research references, both QC layers, content review, editing, and export. Briefly explain the tools and technologies used.

Evaluation Criteria

- Workflow functionality and technical implementation.
- Research quality and factual accuracy.
- Effectiveness of both QC layers.
- Instructional quality and learning objective alignment.
- Problem-solving approach and technical decisions.

Important Notes

- Plagiarism is strictly not accepted. Acknowledge external resources used.
- AI tools are allowed, but you must be able to explain your implementation.
- This task is solely for hiring evaluation purposes. The submission and its output will not be used for internal or commercial purposes.
- Deployment is not required.



Submission Guidelines

The submission form can be submitted only once. Please ensure that you are submitting the correct project artifacts, links, and information. Resubmissions will not be permitted.

Submission Link